Computer Vision —— Object Recognition

2026/9/22

桑 农

Slides from CV course with R. Fergus, and S. Savarese, S. Lazebnik, D. Lowe, P. Viola

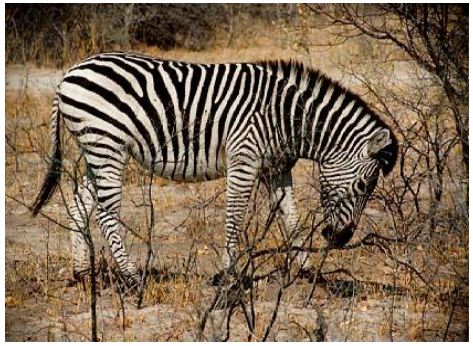
Agenda

- Introduction
- Objects & Faces recognition
- Bag-of-words models
- Part-based models

Timeline of recognition

- 1965-late 1980s: alignment, geometric primitives
- Early 1990s: invariants, appearance-based methods
- Mid-late 1990s: sliding window approaches
- Late 1990s: feature-based methods
- Early 2000s – present : parts-and-shape models
- 2003 – present: bags of features
- Present trends: combination of local and global methods, modeling context, integrating recognition and segmentation

Object categorization: the statistical viewpoint



$$p(zebra \mid image)$$

vs.

$$p(no\ zebra \mid image)$$

- Bayes rule:

$$\underbrace{\frac{p(zebra \mid image)}{p(no\ zebra \mid image)}}_{\text{posterior ratio}} = \underbrace{\frac{p(image \mid zebra)}{p(image \mid no\ zebra)}}_{\text{likelihood ratio}} \cdot \underbrace{\frac{p(zebra)}{p(no\ zebra)}}_{\text{prior ratio}}$$

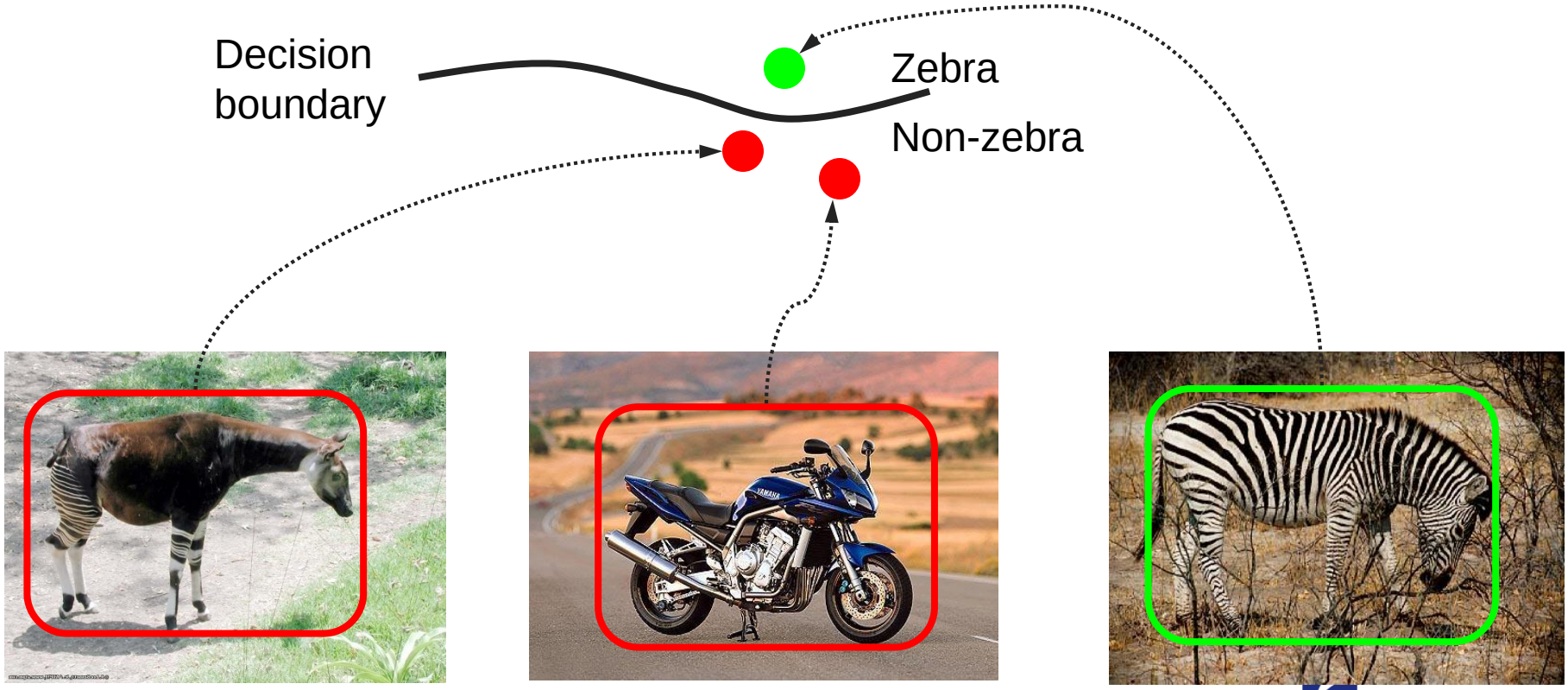
Object categorization: the statistical viewpoint

$$\underbrace{\frac{p(zebra \mid image)}{p(no\ zebra \mid image)}}_{\text{posterior ratio}} = \underbrace{\frac{p(image \mid zebra)}{p(image \mid no\ zebra)}}_{\text{likelihood ratio}} \cdot \underbrace{\frac{p(zebra)}{p(no\ zebra)}}_{\text{prior ratio}}$$

- **Discriminative methods model posterior**
- **Generative methods model likelihood and prior**

Discriminative

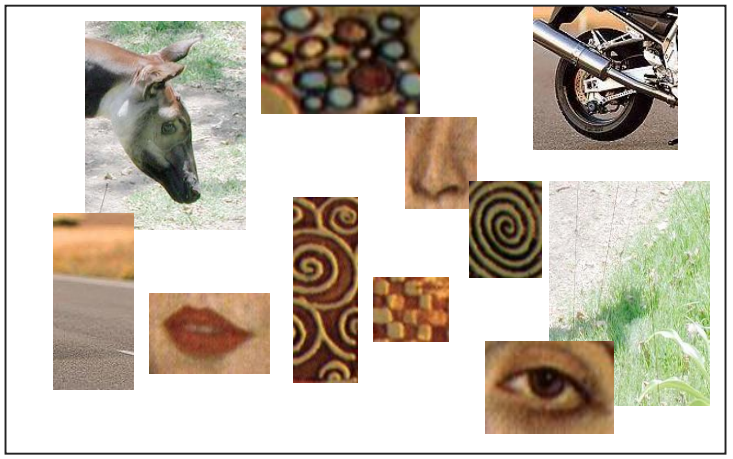
- Direct modeling of $\dfrac{p(zebra \mid image)}{p(no\ zebra \mid image)}$



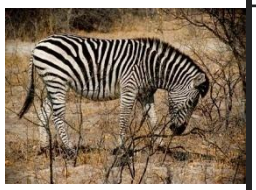
Generative

- Model $p(image \mid zebra)$ and $p(image \mid no\ zebra)$









$p(image \mid zebra)$	$p(image \mid no\ zebra)$
Low	Middle
High	Middle→Low

Three main issues

- Representation

  How to represent an object category

- Learning

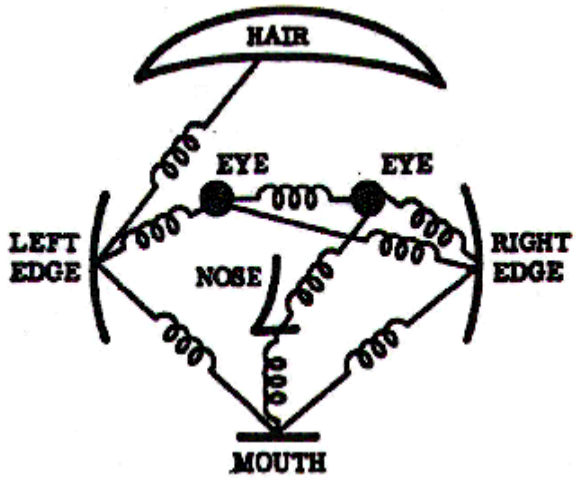
  How to form the classifier, given training data

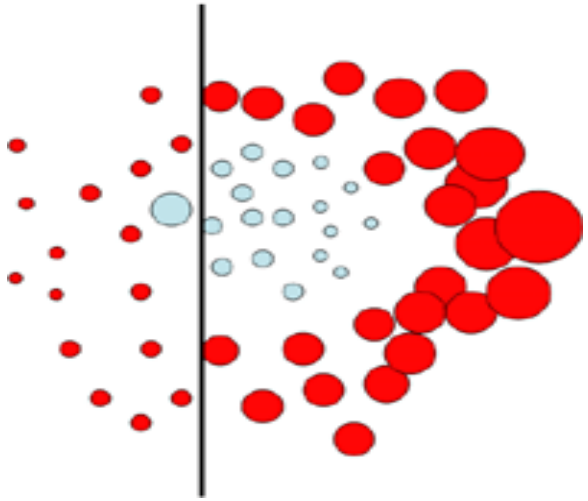
- Recognition

  How the classifier is to be used on novel data

Representation

Generative / discriminative / hybrid

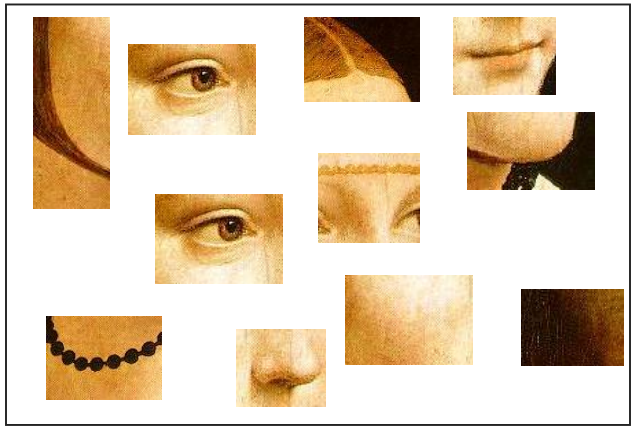


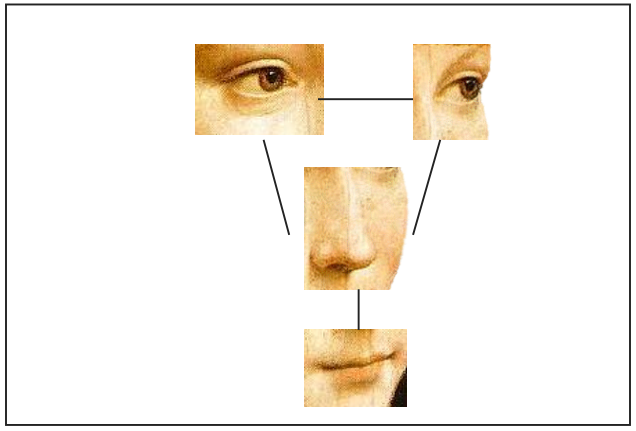


Representation

Generative / discriminative / hybrid

Appearance only or location and appearance





Representation

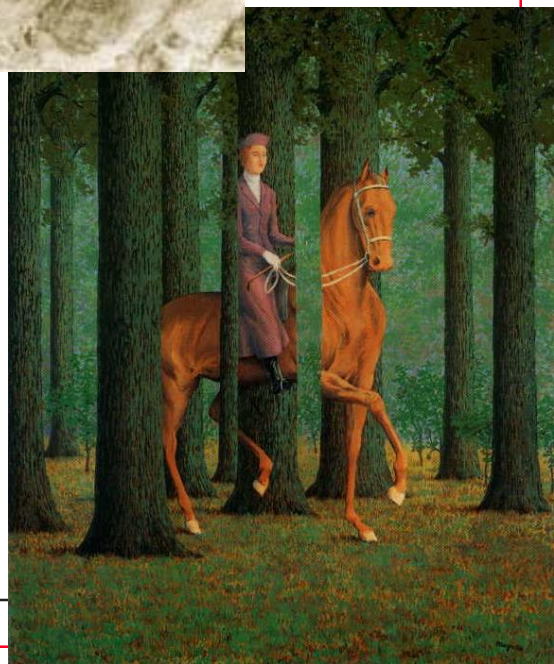
Generative / discriminative / hybrid

Appearance only or location and appearance

Invariances

- View point
- Illumination
- Occlusion
- Scale
- Deformation
- Clutter
- etc.





Representation

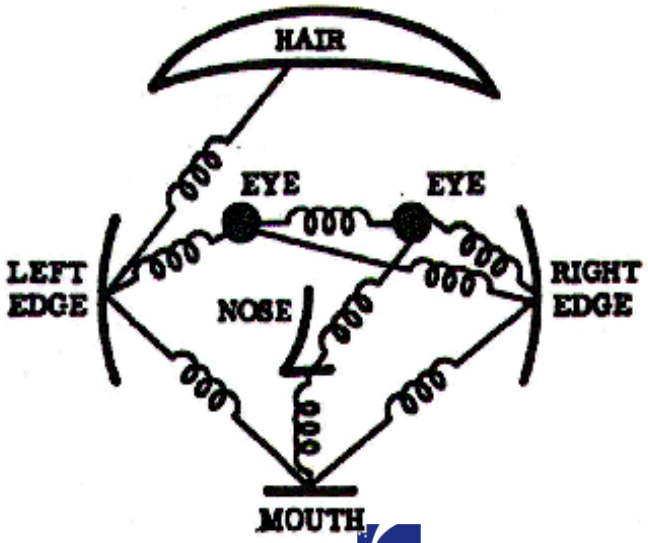
Generative / discriminative / hybrid

Appearance only or location and appearance

invariances

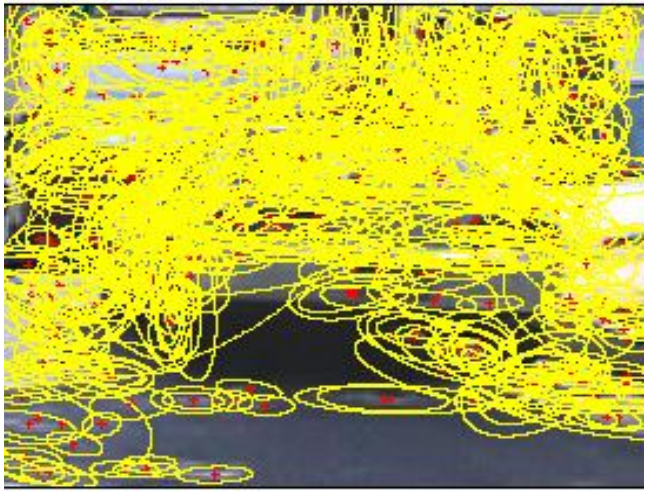
Part-based or global w/sub-window





Representation

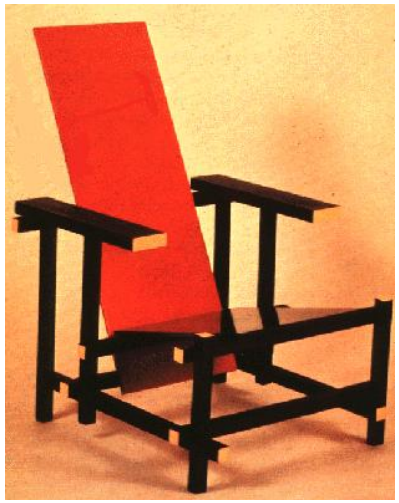
- Generative / discriminative / hybrid
- Appearance only or location and appearance
- invariances
- Parts or global w/sub-window
- Use set of features or each pixel in image





Learning

Unclear how to model categories, so we learn what distinguishes them rather than manually specify the difference -- hence current interest in machine learning



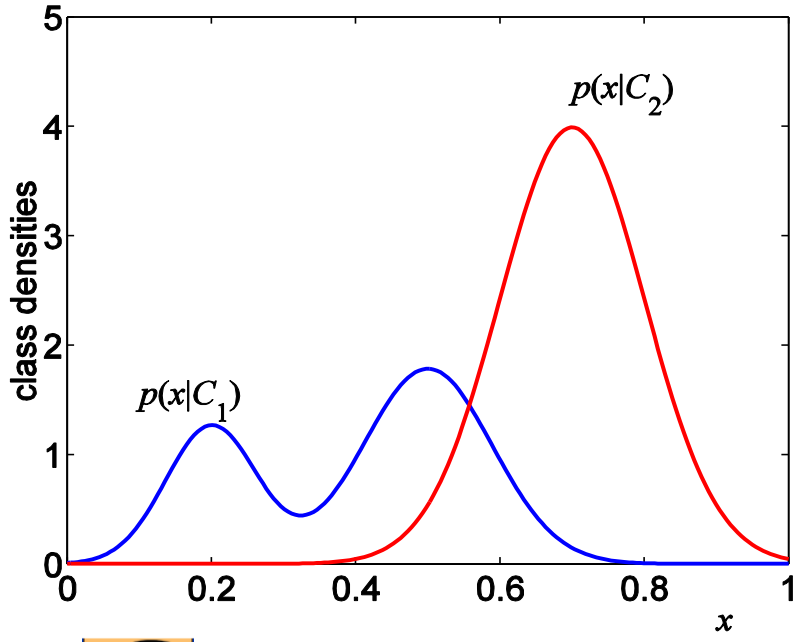


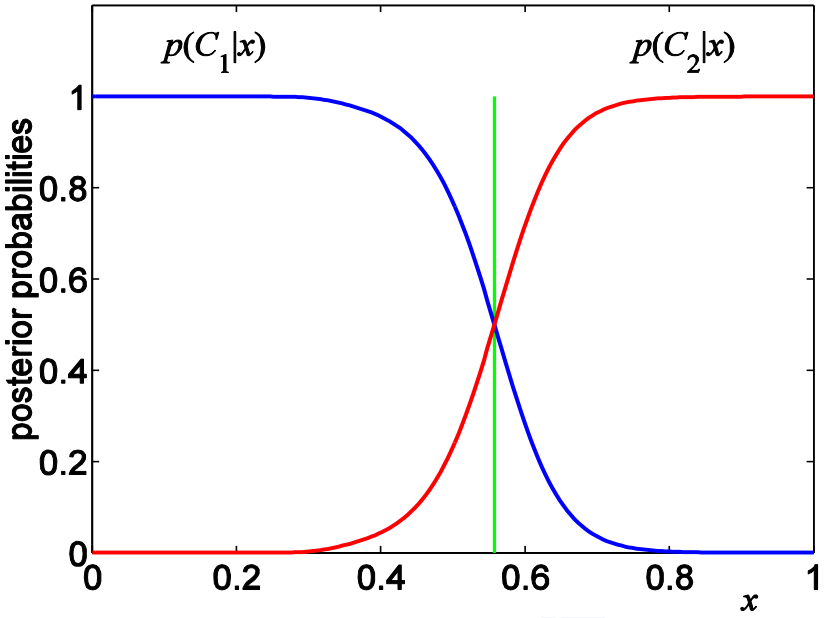


Learning

Unclear how to model categories, so we learn what distinguishes them rather than manually specify the difference -- hence current interest in machine learning)

Methods of training: generative vs. discriminative





Learning

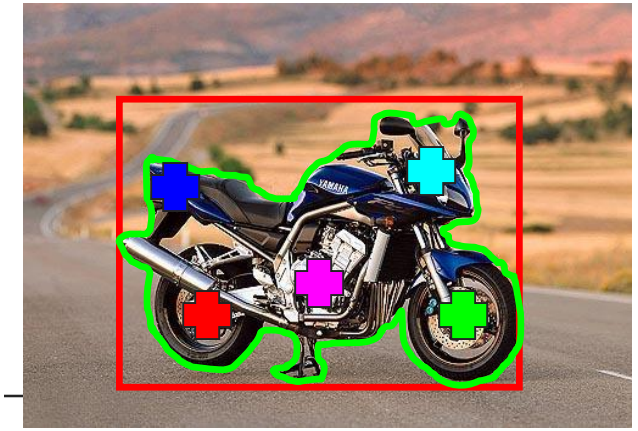
Unclear how to model categories, so we learn what distinguishes them rather than manually specify the difference -- hence current interest in machine learning)

What are you maximizing? Likelihood (Gen.) or performances on train/validation set (Disc.)

Level of supervision

Manual segmentation; bounding box; image labels; parts labels

Contains a motorbike



Learning

Unclear how to model categories, so we learn what distinguishes them rather than manually specify the difference -- hence current interest in machine learning)

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Batch/incremental (on category and image level; user-feedback)

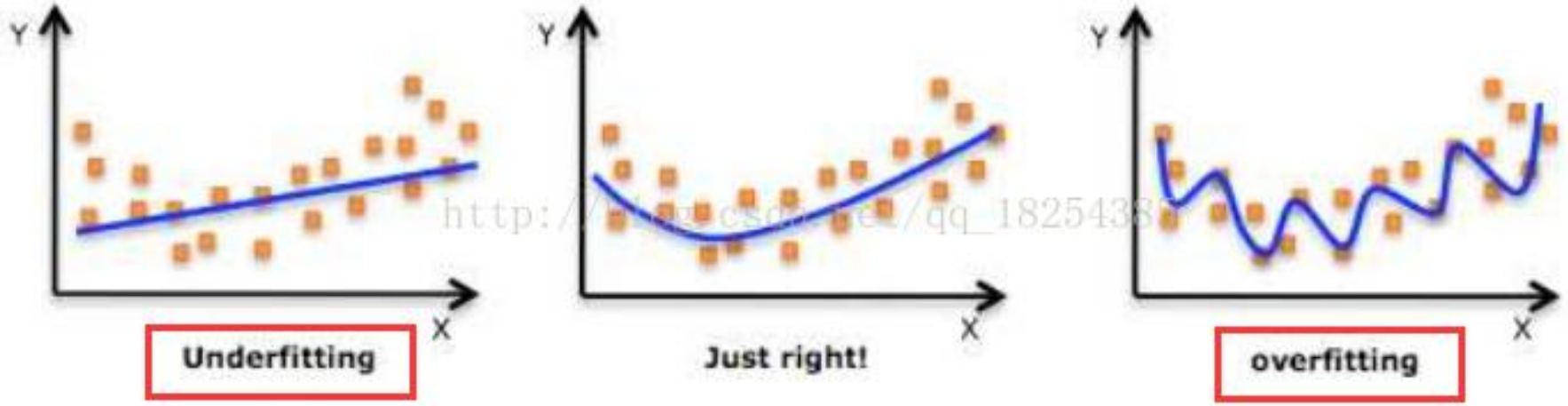


Learning

- Unclear how to model categories, so we learn what distinguishes them rather than manually specify the difference -- hence current interest in machine learning)
- What are you maximizing? Likelihood (Gen.) or performances on train/validation set (Disc.)
- Level of supervision
  - Manual segmentation; bounding box; image labels; parts labels
- Batch/incremental (on category and image level; user-feedback)
- Training images:
  - Issue of overfitting

Learning

Unclear how to model categories, so we learn what distinguishes them rather than manually



Training images:

Issue of overfitting

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Training images:

- Issue of overfitting
- Negative images for discriminative methods

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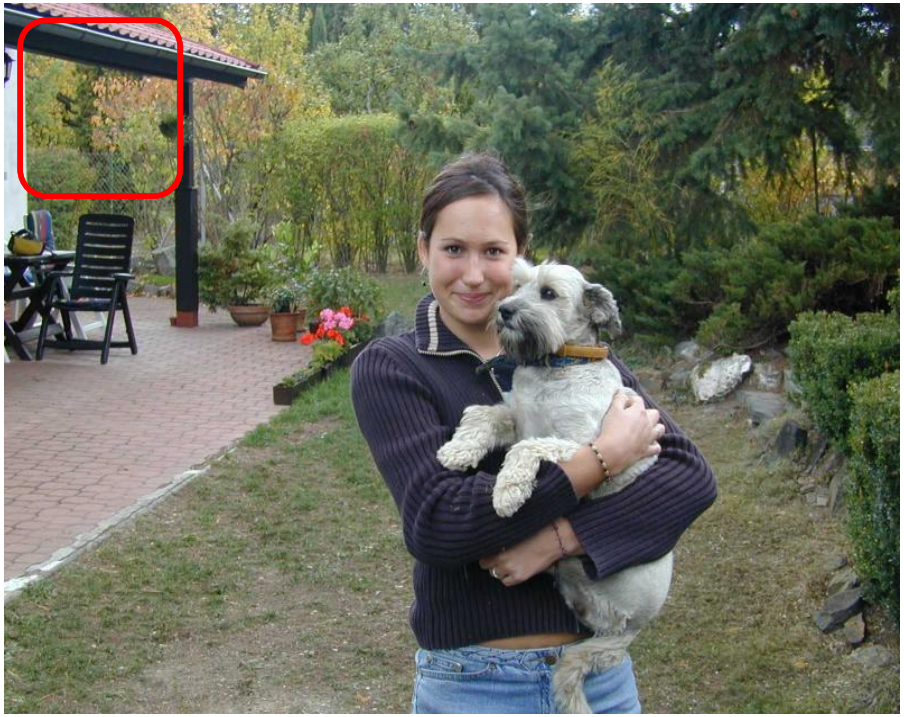
- Issue of overfitting
- Negative images for discriminative methods

Priors

Recognition

Scale / orientation range to search over

Speed



Agenda

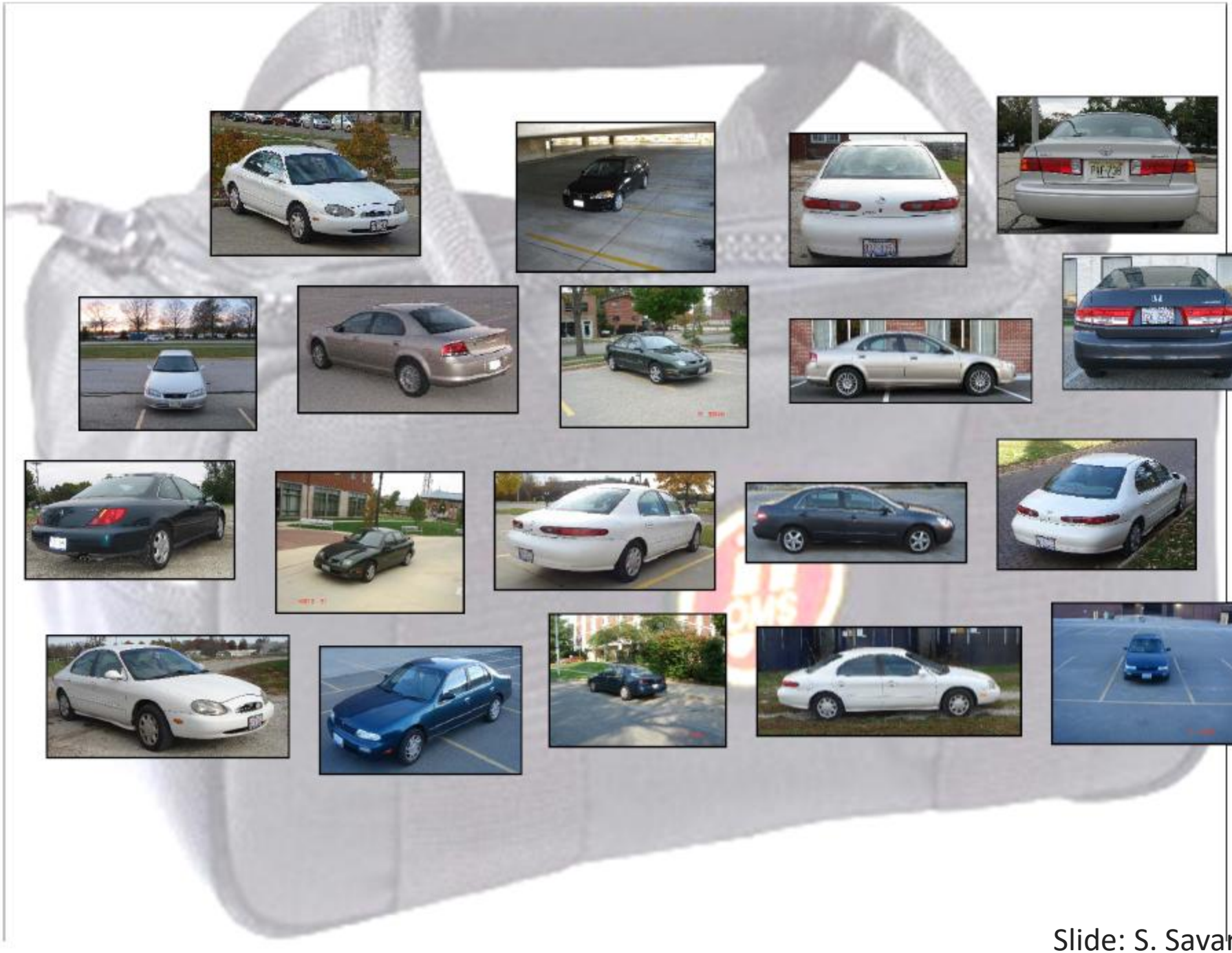
- Introduction
- Objects & Faces recognition
- Bag-of-words models
- Part-based models

Overview

- Specific Object Recognition
  - Lowe‘04
  - Rothganger et al.‘03
- Face Detection
  - Sub-space methods
  - Neural Network-based approaches
  - Viola & Jones
- Face Recognition

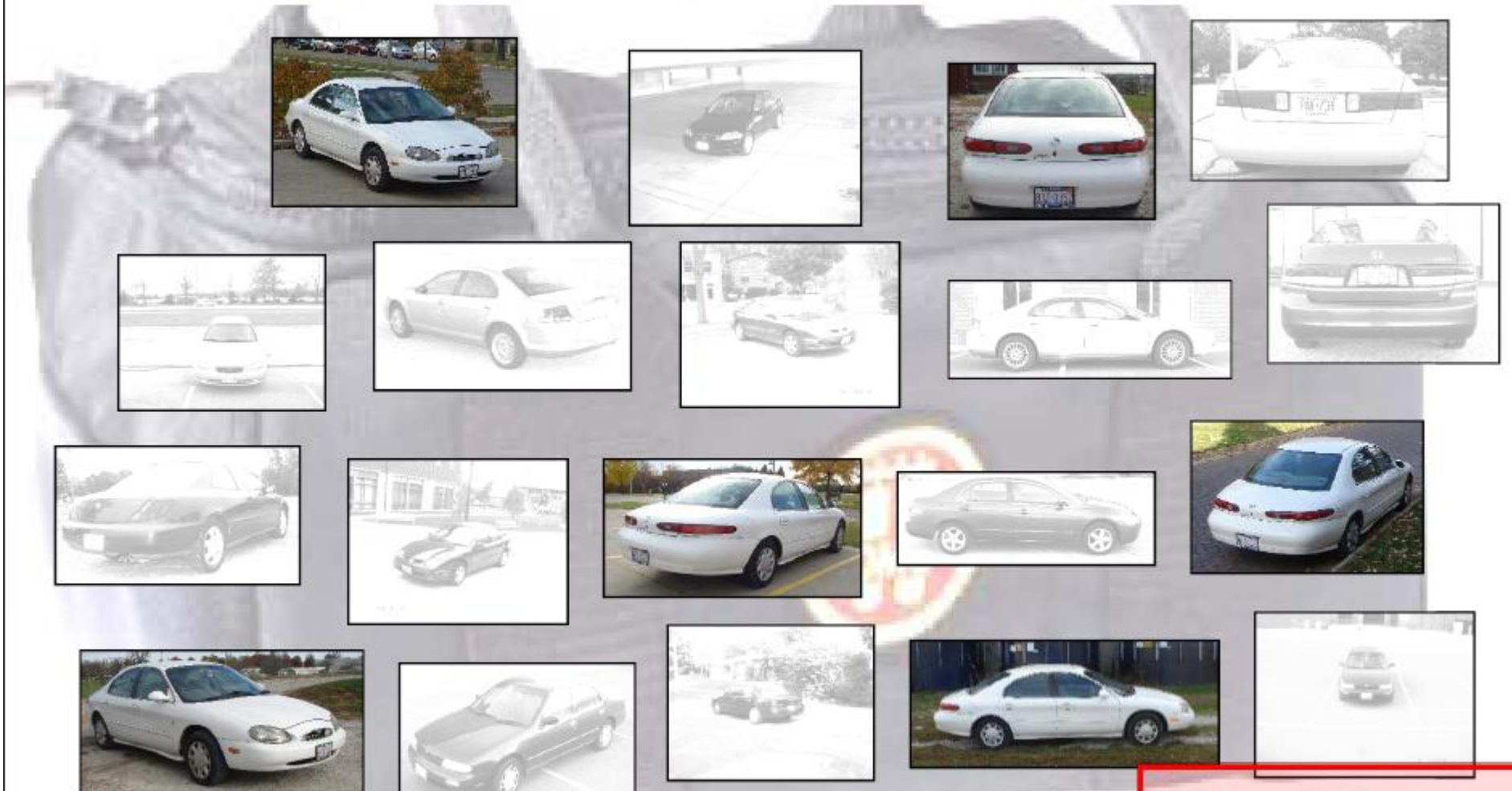
Specific Object Recognition

Slides from S. Savarese, D. Lowe and F. Rothganger



Slide: S. Savarese

Single 3D object recognition



- Ballard, '81
- Grimson & L.-Perez, '87
- Lowe, '87

- Edelman et al. '91
- Ullman & Barsi, '91
- Rothwell '92
- Linderberg, '94
- Murase & Nayar '94

- Zhang et al '95
- Schmid & Mohr, '96
- Schiele & Crowley, '96
- **Lowe, '99**
- **Jacob & Barsi, '99**
- Mahamud and Herbert, 00

- **Rothganger et al., '04**
- **Ferrari et al, '05**
- Moreels and Perona, 05
- **Brown & Lowe '05**
- **Snavely et al '06**
- **Yin & Collins, '07**

Where is the crunchy nut?





Challenges:

- Variability due to:

  View point

  Illumination

  Occlusions

  But not intra-class variation

Recognition of single 3D objects

- Rothganger et al. ’04, ’06
- Brown et al, ‘05
- Lowe ‘99, ‘04
- Ferrari et al. ’04, ‘06
- Lazebnick et al ’04

-Representation

-Features

-2D/3D Geometrical constraints

-Model learning

-Recognition

-Hypothesis generation

-Validation

Representation

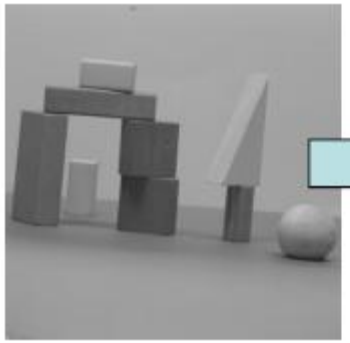
Interest points -- or Regions (group of interest points)

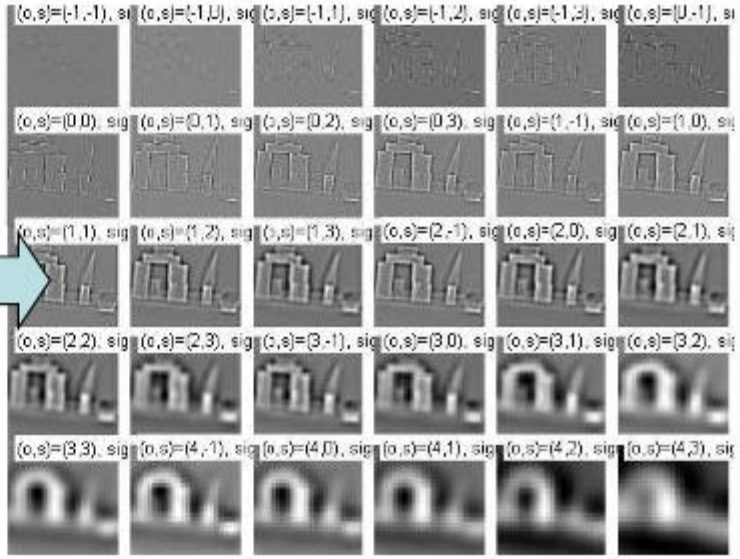
- Detection
  - Difference of Gaussian (DOG) [Lowe '99]
  - Harris-Laplacian [Mikolajczyk & Schmid '01]
  - Kadir-Brady [Kadir et al. '01]
  - Laplacian [Gårding & Lindeberg, '96]
- Adaptation [invariants]
  - Scale, rotation
  - Affine
- Description
  - SIFT
  - Color histograms

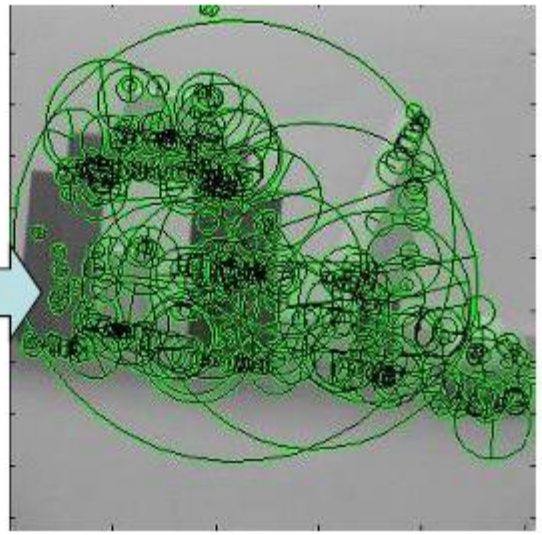
Geometrical constraints

- 2D spatial layout of keypoints
- Tracks of keypoints (regions) across views
- 3D locations and/or surface normals

Difference of Gaussian (DOG): used in Lowe 99, Brown et al '05







Courtesy of D. Lowe

Harris-Laplace: used in Rothganger et al. '06

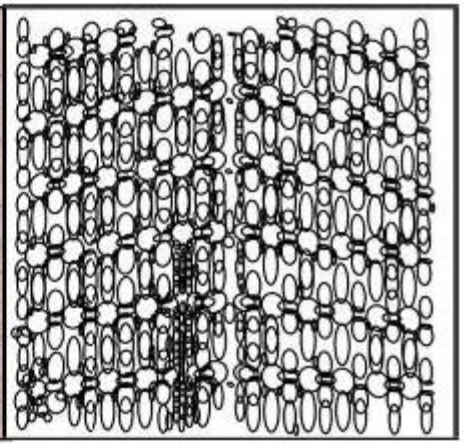




Courtesy of Rothganger et al

Laplacian: used in Lazebnik et al. '04





Representation

Interest points -- or Regions (group of interest points)

- Detection
  - Difference of Gaussian (DOG) [Lowe '99]
  - Harris-Laplacian [Mikolajczyk & Schmid '01]
  - Kadir-Brady [Kadir et al. '01]
  - Laplacian [Gårding & Lindeberg, '96]

→

- x,y
- Scale
- Orientation
- Affine structure

- Adaptation [invariants]
  - Scale, rotation
  - Affine

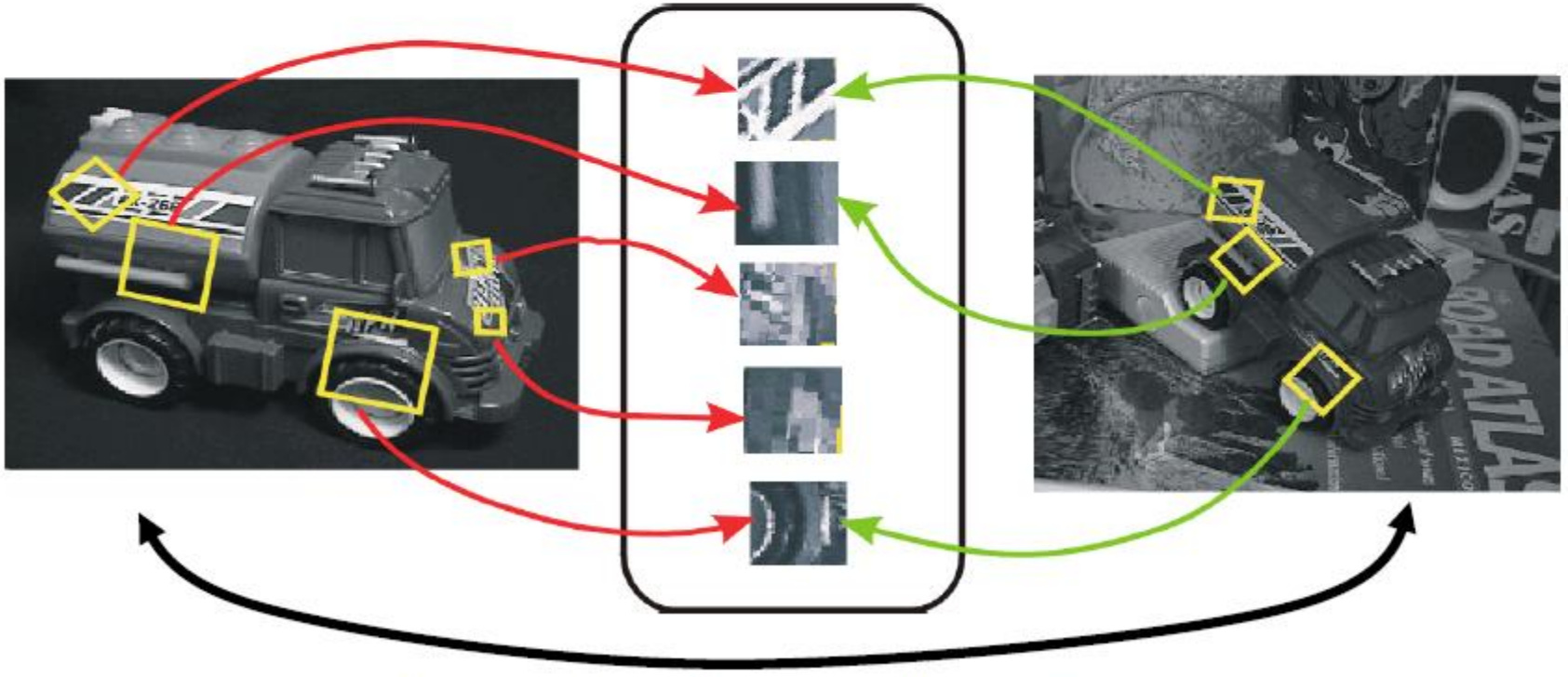
- Description
  - SIFT
  - Color histograms

Geometrical constraints

- 2D spatial layout of keypoints
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Adaptation

- keypoints are transformed in order to be invariant to translation, rotation, scale, and other geometrical parameters



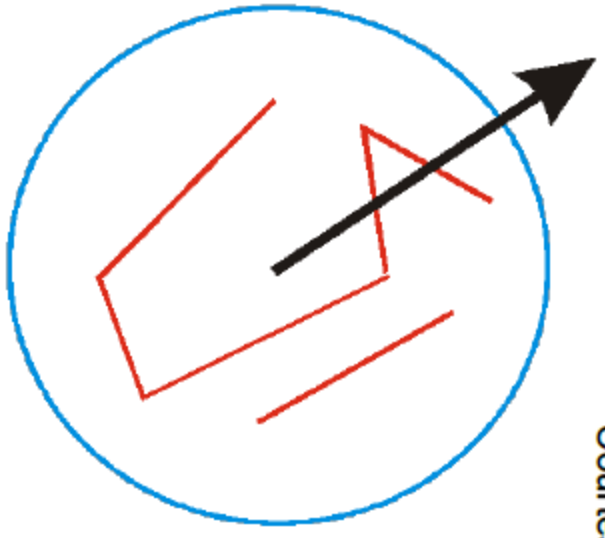
Courtesy of D. Lowe

Change of scale, pose, illumination…

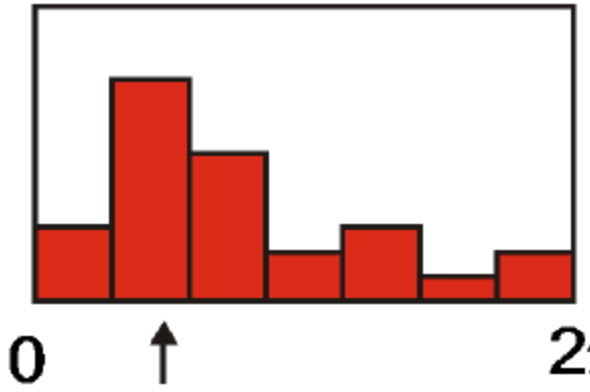
Scale & orientation adaptation

[used in Lowe '99]

detector → • x,y
•Scale
•Orientation



- SIFT: Create histogram of local gradient directions computed at selected scale
- Assign canonical orientation at peak of smoothed histogram

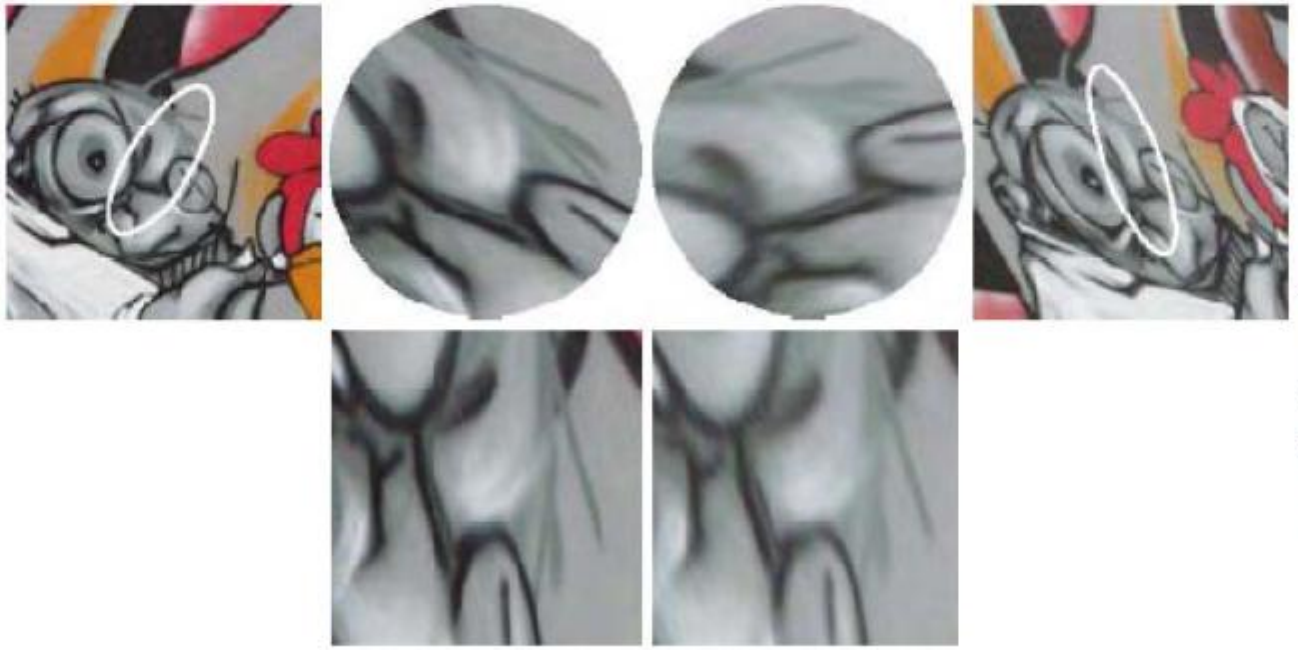


Courtesy of D. Lowe

Affine adaptation

[used in Rothganger et al. '03, '06]
[Lazebnick et al '04]

1. Define elliptical region **using second moment matrix**
2. Use main canonical orientation to remove orientation ambiguity
3. Map ellipsis onto unit square



Courtesy of Rothganger et al

Representation

Interest points -- or Regions (group of interest points)

- Detection
  - Difference of Gaussian (DOG) [Lowe '99]
  - Harris-Laplacian [Mikolajczyk & Schmid '01]
  - Kadir-Brady [Kadir et al. '01]
  - Laplacian [Gårding & Lindeberg, '96]



- x,y
- Scale
- Orientation
- Affine structure

- Adaptation [invariants]
  - Scale, rotation
  - Affine
- Description
  - SIFT
  - Color histograms

Geometrical constraints

- 2D spatial layout of keypoints
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Keypoint description

[Lowe '99]

- Thresholded image gradients are sampled over 16x16 array of locations in scale space
- Create array of orientation histograms
- 8 orientations x 4x4 histogram array = 128 dimensions

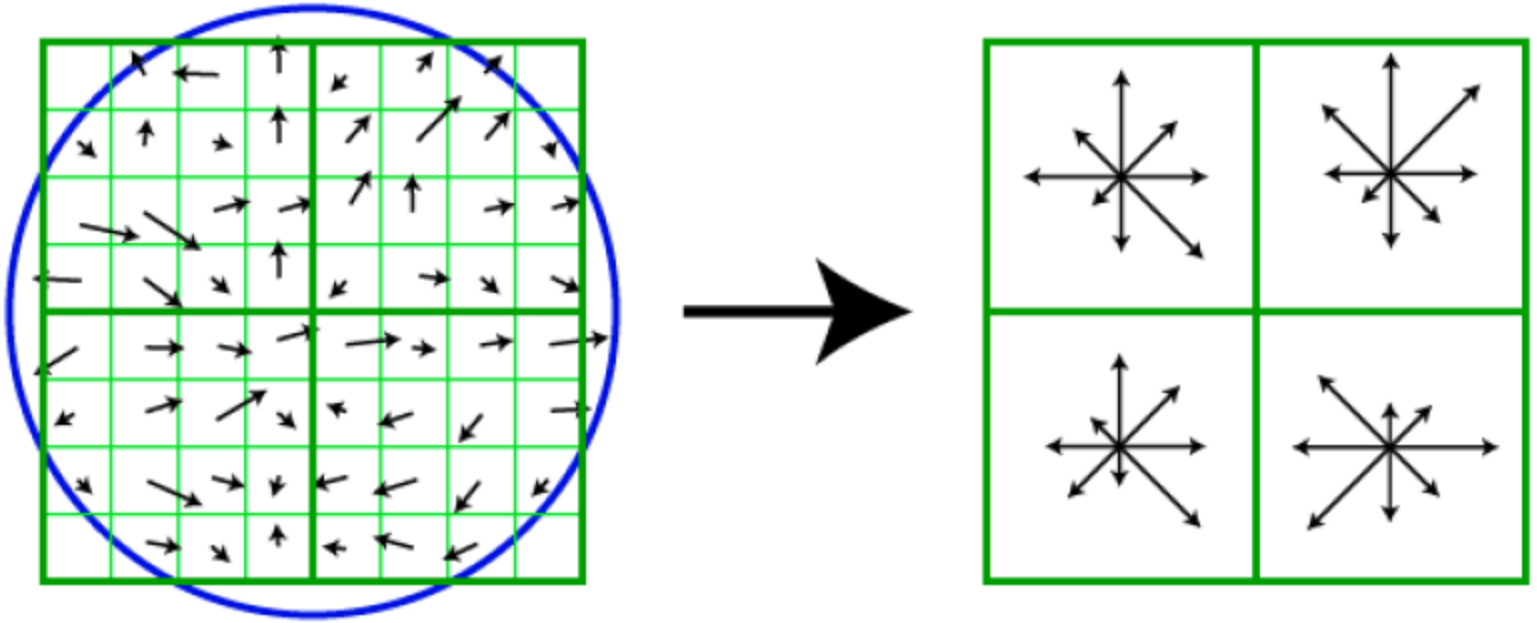


Image gradients

Keypoint descriptor

Representation

Interest points -- or Regions (group of interest points)

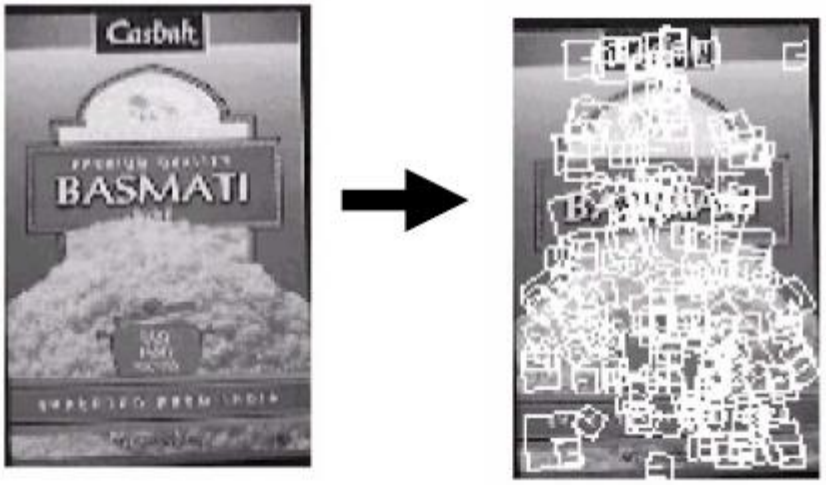
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Geometrical constraints

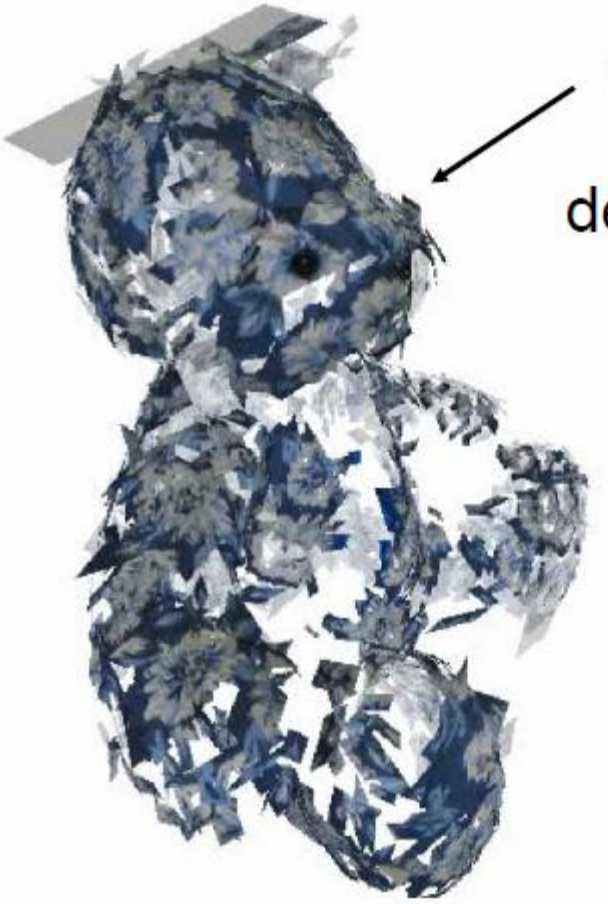
- 3D locations and/or surface normals
- 2D spatial layout of keypoints [collections of views]
- Tracks of keypoints (regions) across views

Object representation: 2D or 3D location of key points

[Lowe '99]



Rothganger et al. '06



Courtesy of Rothganger et al

Basic scheme

-Representation

-Features

-2D/3D Geometrical constraints

-Model learning

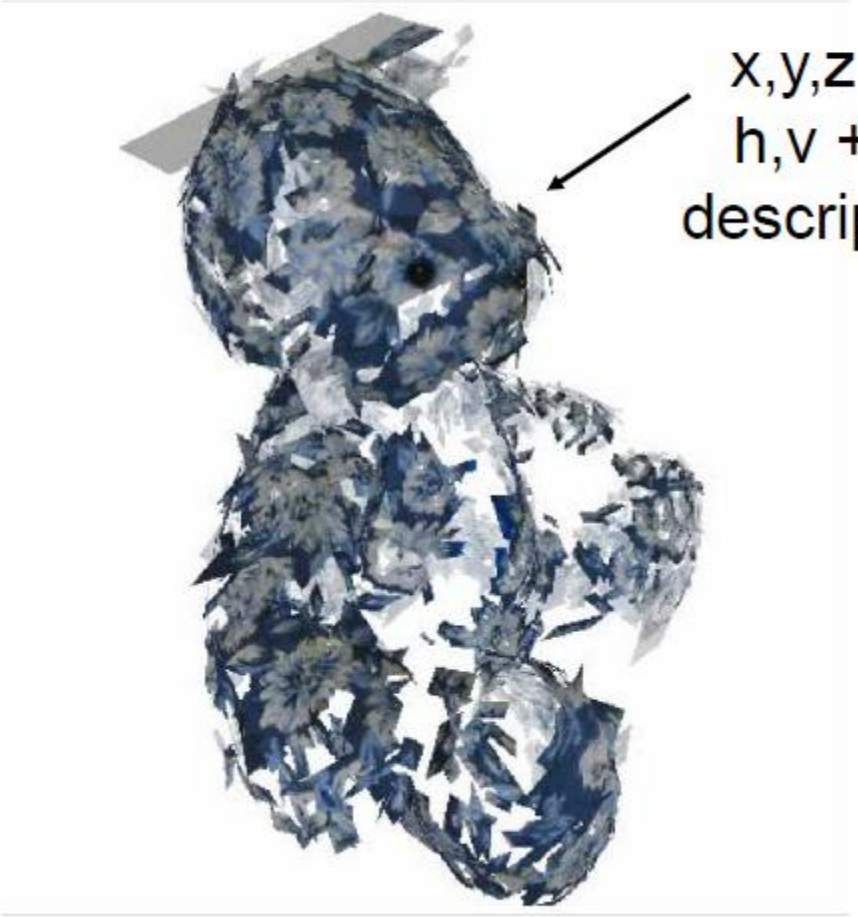
-Recognition

-hypothesis generation

-validation

Model learning

Rothganger et al. '03 '06



Courtesy of Rothganger et al

Model learning

Rothganger et al. '03 '06

Build a 3D model:



- N images of object from N different view points

- Match key points between consecutive views
[create sample set]



- Use affine structure from motion to compute 3D location and orientation + camera locations

- Affine factorization Tomasi & Kanade '92
- RANSAC
- 2 matches are needed rather than 4 thanks to affine invariant patches

Model learning

Rothganger et al. '03 '06

Build a 3D model:



- N images of object from N different view points

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Use affine structure from motion to compute 3D location and orientation + camera locations

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- RANSAC
- 2 matches are needed rather than 4 thanks to affine invariant patches

- Find connected components
- Use bundle adjustment to refine model
- Upgrade model to Euclidean assuming zero skew and square pixels

$$E = \sum_{j=1}^{n} \sum_{i \in I_j} |\mathcal{S}_{ij} - \mathcal{M}_i \mathcal{N}_j|^2,$$

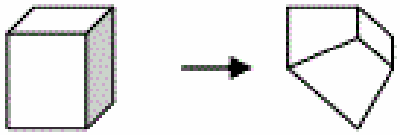
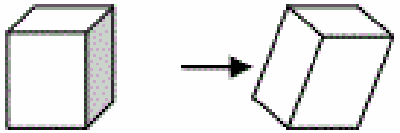
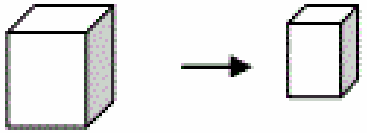
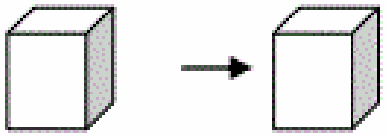
变换	自由度	变换矩阵	形变	不变量
射影变换	15	$\begin{bmatrix} p_{11} & p_{12} & p_{13} & p_{14} \\ p_{21} & p_{22} & p_{23} & p_{24} \\ p_{31} & p_{32} & p_{33} & p_{34} \\ p_{41} & p_{42} & p_{43} & p_{44} \end{bmatrix}$		交比
仿射变换	12	$\begin{bmatrix} a_{11} & a_{12} & a_{13} & a_{14} \\ a_{21} & a_{22} & a_{23} & a_{24} \\ a_{31} & a_{32} & a_{33} & a_{34} \\ 0 & 0 & 0 & 1 \end{bmatrix}$		相对距离 平行性 无穷远平面
度量变换	7	$\begin{bmatrix} \sigma r_{11} & \sigma r_{12} & \sigma r_{13} & t_x \\ \sigma r_{21} & \sigma r_{22} & \sigma r_{23} & t_y \\ \sigma r_{31} & \sigma r_{32} & \sigma r_{33} & t_z \\ 0 & 0 & 0 & 1 \end{bmatrix}$		相对距离 夹角 绝对二次曲线
欧氏变换	6	$\begin{bmatrix} r_{11} & r_{12} & r_{13} & t_x \\ r_{21} & r_{22} & r_{23} & t_y \\ r_{31} & r_{32} & r_{33} & t_z \\ 0 & 0 & 0 & 1 \end{bmatrix}$		绝对距离

图 5　三维几何空间层次的变换

Learnt models

Rothganger et al. ‘03 ’06

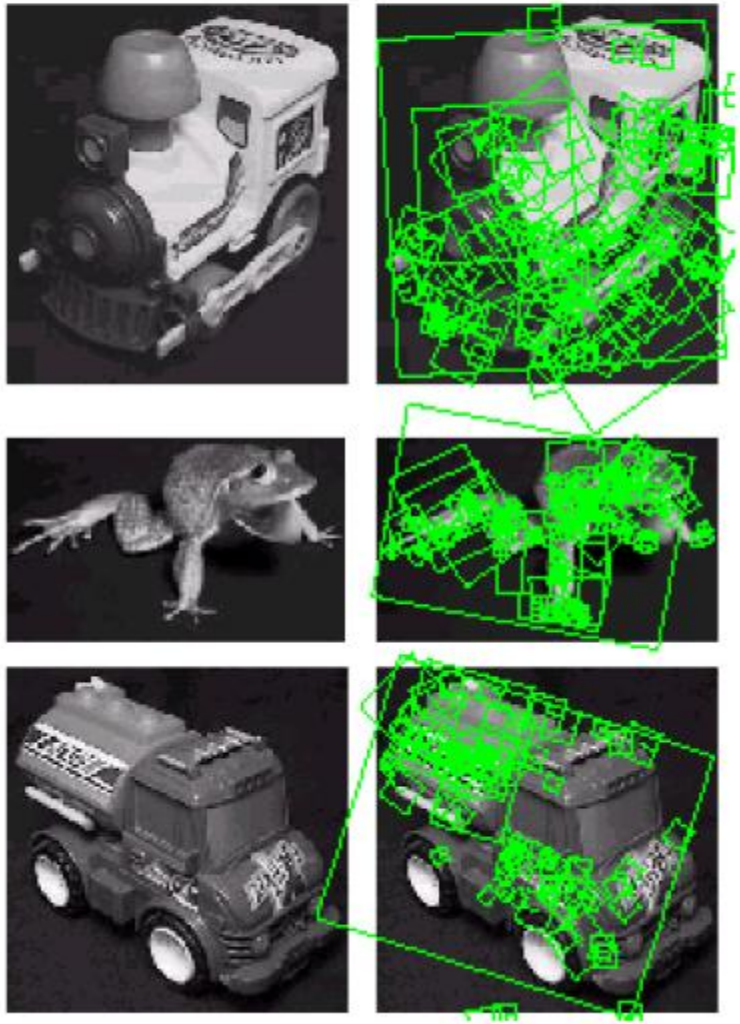




Courtesy of Rothganger et al

Learnt models

[Lowe '99]



Basic scheme

- Representation
  - Features
  - 2D/3D Geometrical constraints
- Model learning
- Recognition [object instance from object model]
  - hypothesis generation
  - Model verification

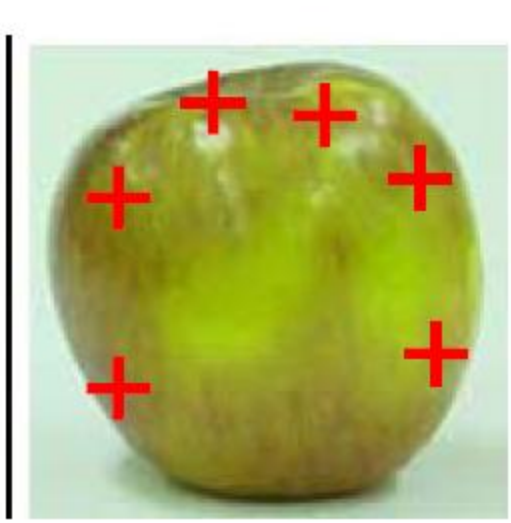
Recognition

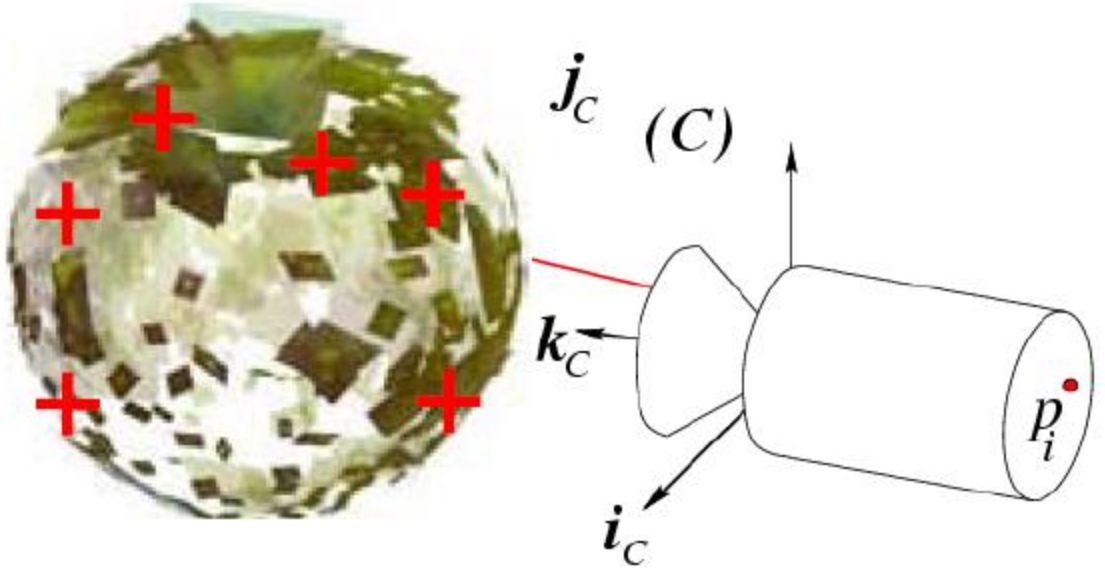
Goal: given a query image I, identify object model in the image I (match learned model to I)

- Generate hypothesis
- Verify hypothesis
- Select hypothesis with lowest fitting error
- Generate recognition results

Recognition

[Rothganger et al. '03 '06]

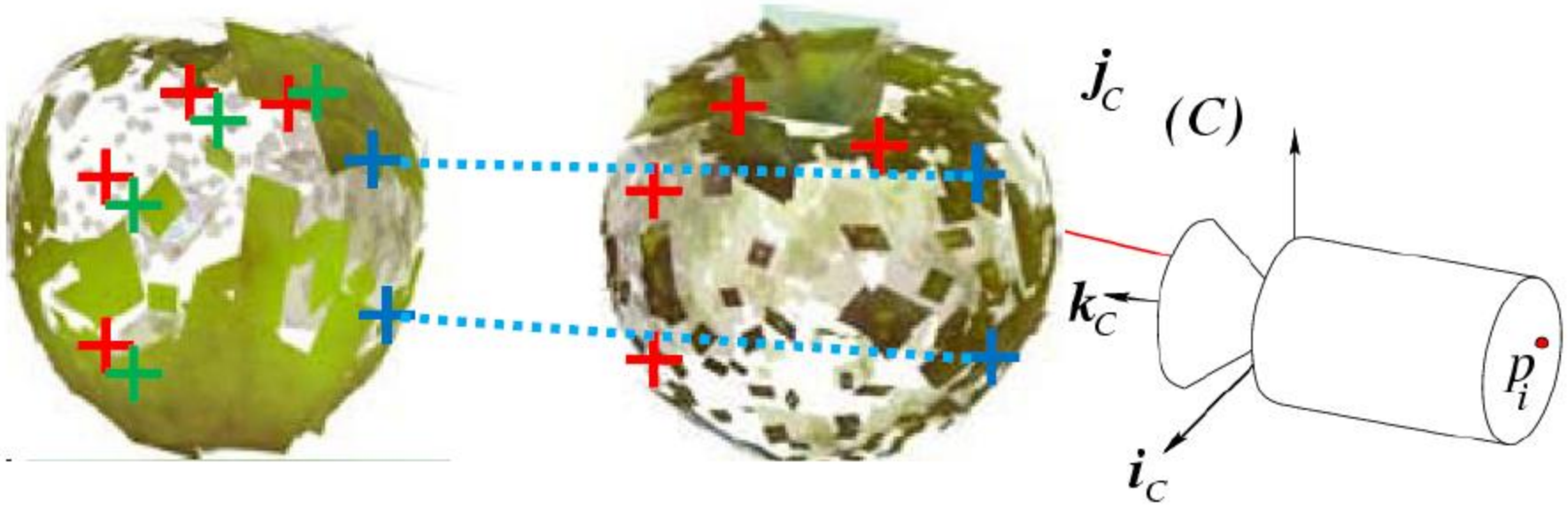




1. Find matches between model and test image features

Recognition

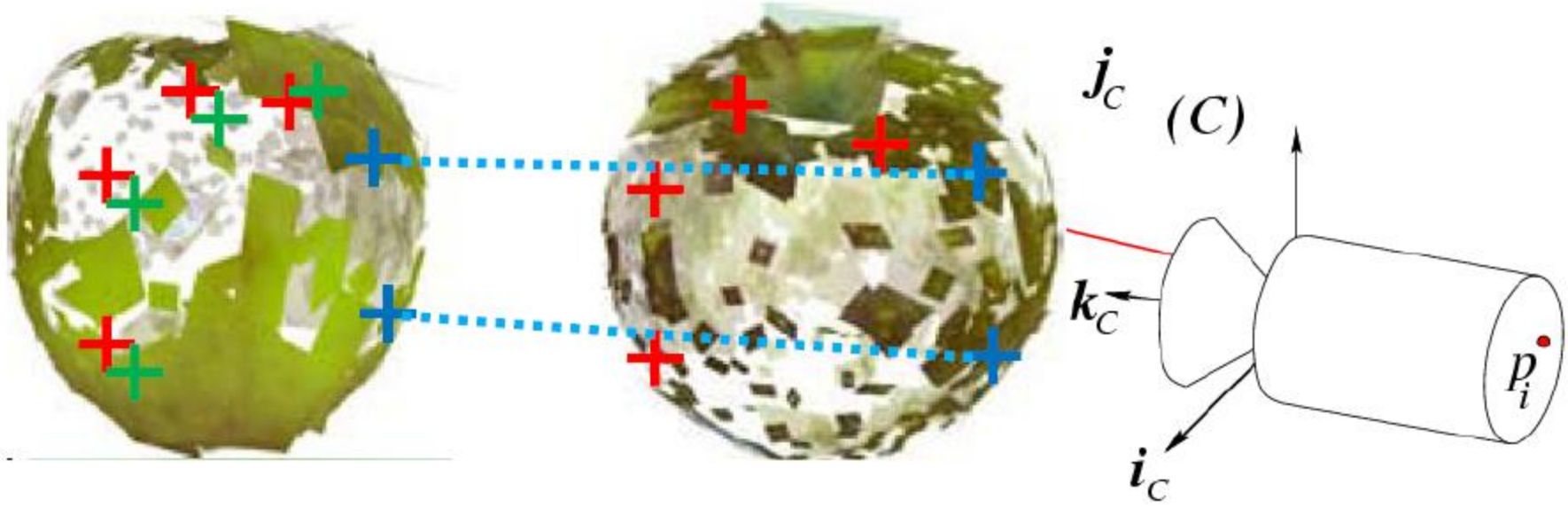
[Rothganger et al. '03 '06]



1. Find matches between model and test image features
2. Generate hypothesis:
   - Compute transformation M from N matches (N=2; affine camera; affine key points)
3. Model verification
   - Use M to project other matched 3D model features into test image
   - Compute residual = D(projections, measurements)

Recognition

[Rothganger et al. '03 '06]



Goal:

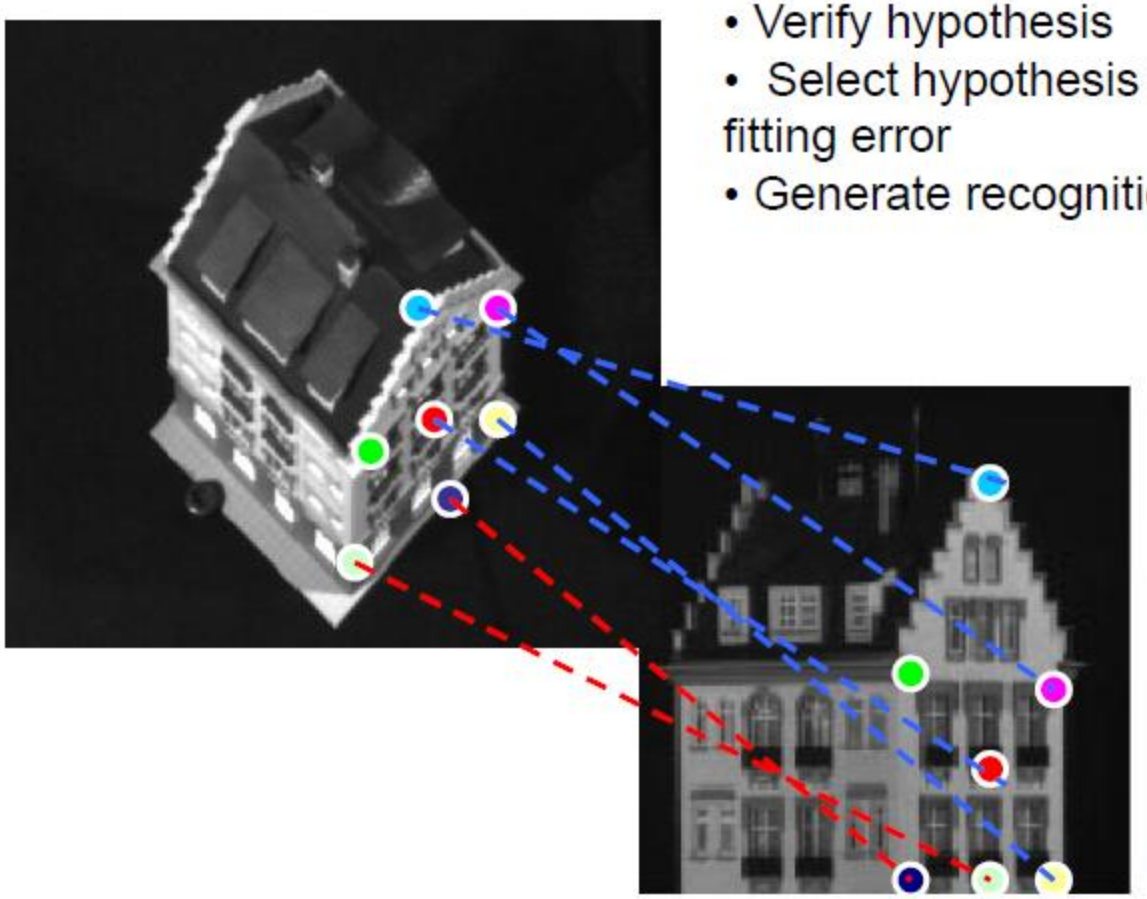
Estimate (fit) the best M in presence of outliers

Recognition

Goal: given a query image I, identify object model in the image I (match learned model to I)

- Generate hypothesis
- Verify hypothesis
- Select hypothesis with lowest fitting error
- Generate recognition results

query



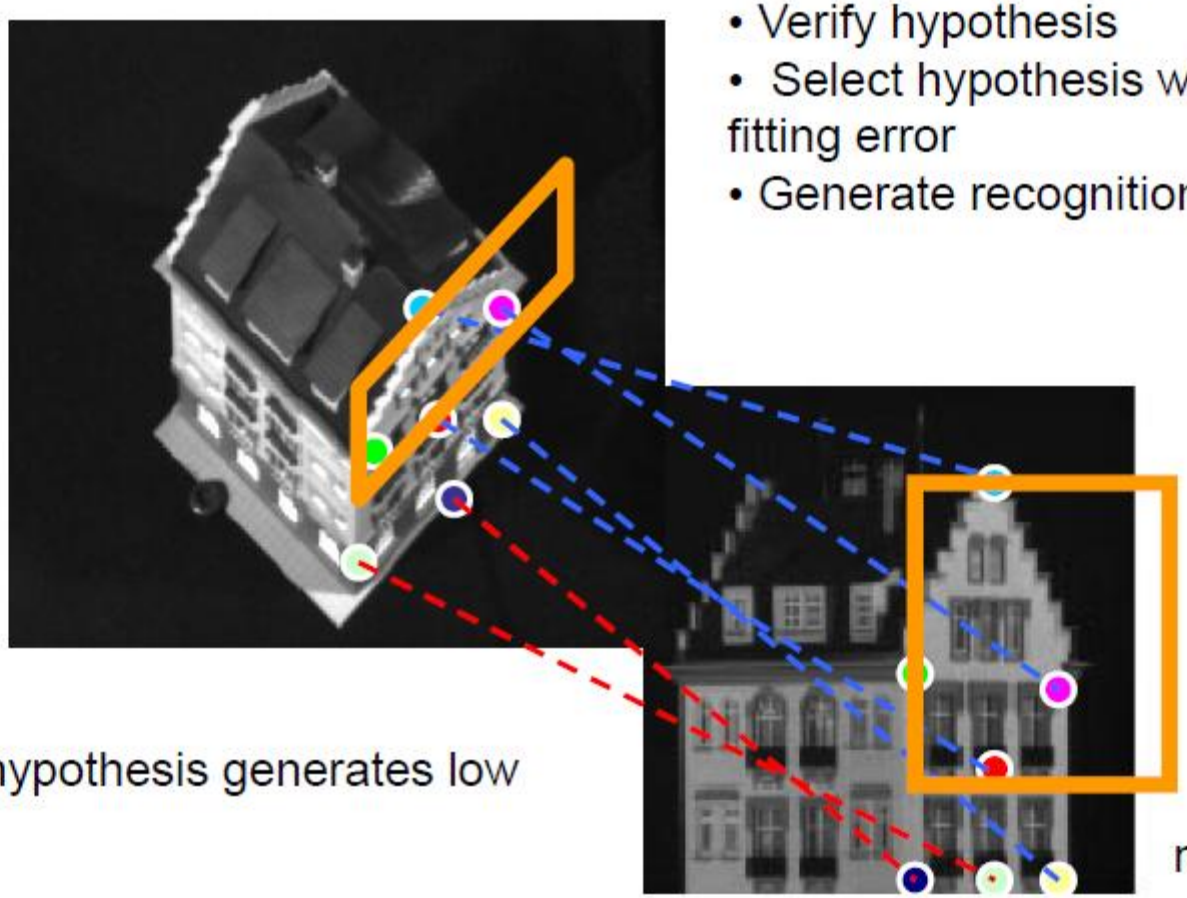
model

Recognition

Goal: given a query image I, identify object model in the image I (match learned model to I)

- Generate hypothesis
- Verify hypothesis
- Select hypothesis with lowest fitting error
- Generate recognition results

query



model

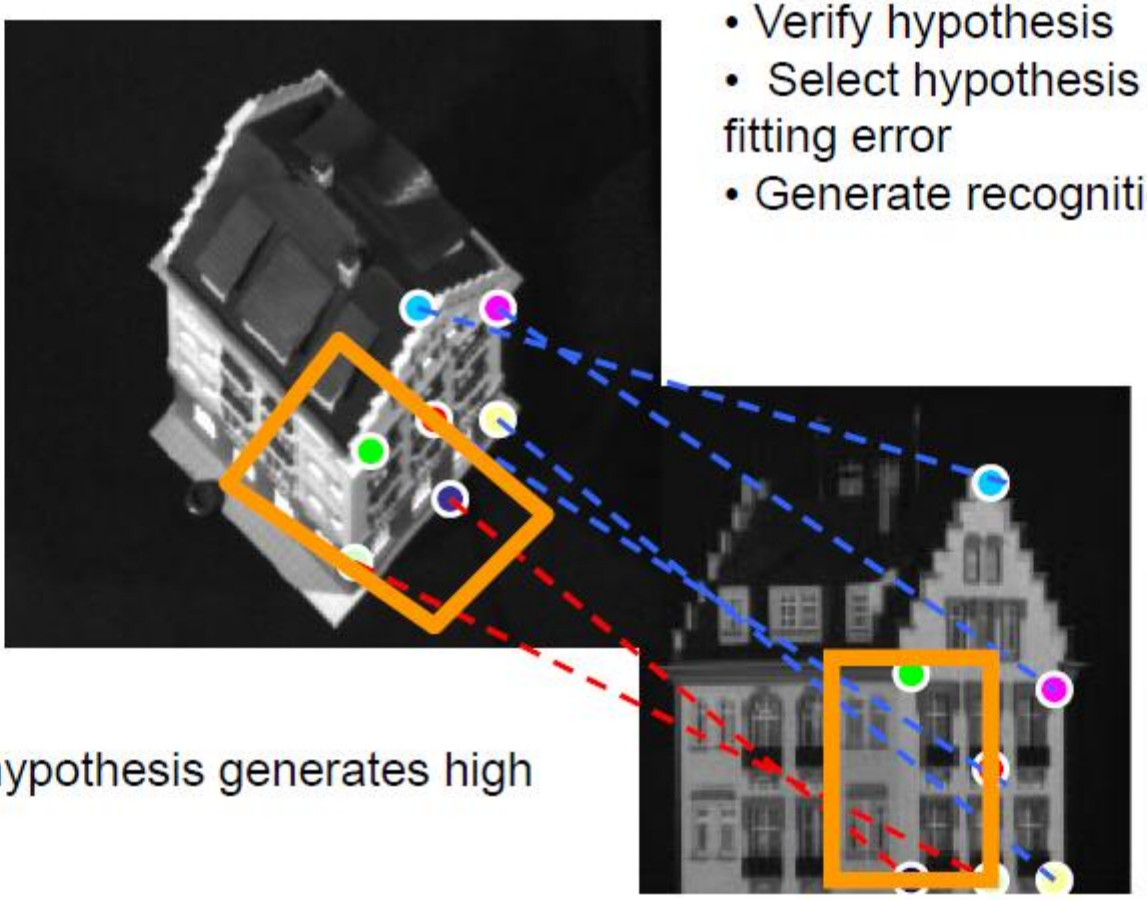
Verification: The hypothesis generates low fitting error

Recognition

Goal: given a query image I, identify object model in the image I (match learned model to I)

- Generate hypothesis
- Verify hypothesis
- Select hypothesis with lowest fitting error
- Generate recognition results

query



model

Verification: The hypothesis generates high fitting error

Object to recognize



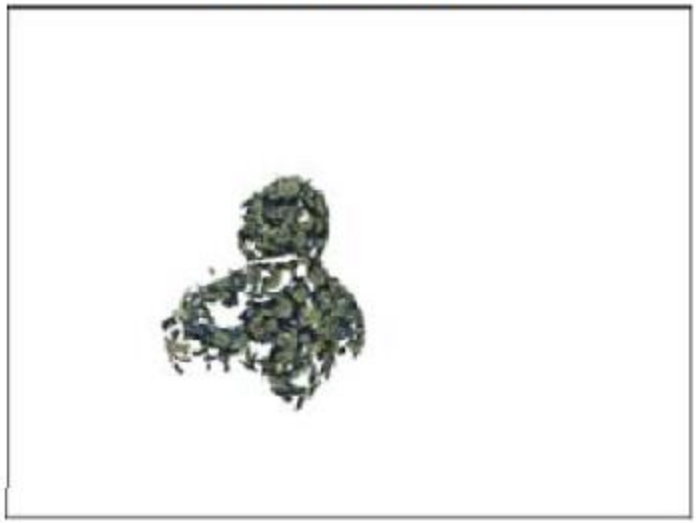
Initial matches based on appearance



Matches verified with geometrical constraints



Recovered pose



Courtesy of Rothganger et al

Basic scheme

-**Representation**
  - -Features
  - -2D/3D Geometrical constraints

-**Model learning**

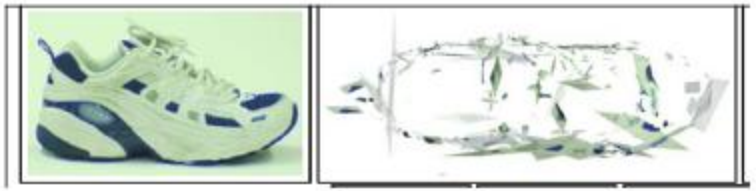
Let's see some results!

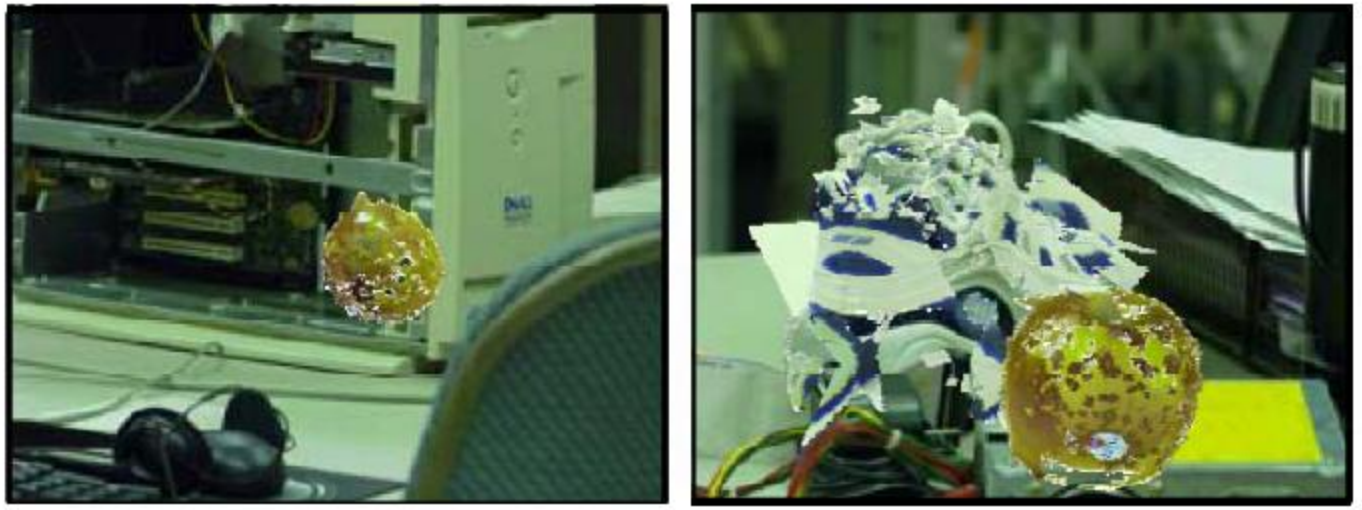
-**Recognition** [object instance from a single image]
  - -hypothesis generation
  - -Model verification

3D Object Recognition results

Rothganger et al. ‘03 ’06







Courtesy of Rothganger et al

•Handle severe clutter

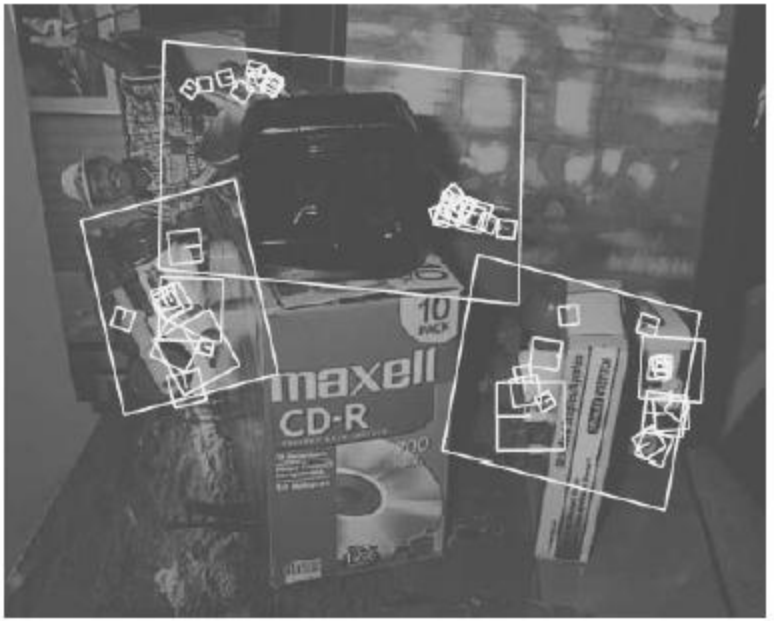
3D Object Recognition results

Lowe. '99, '04







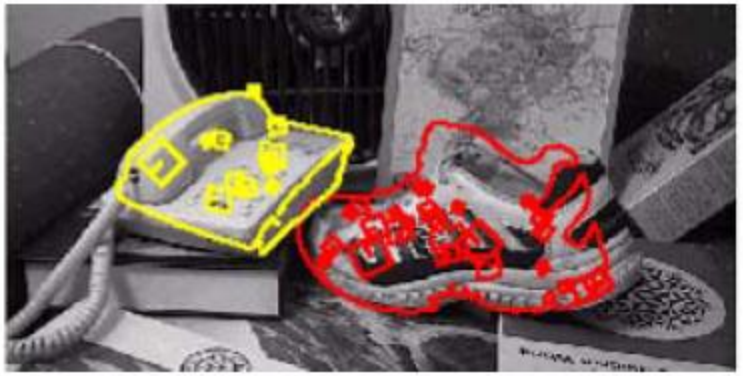












Courtesy of D. Lowe

- Handle severe occlusions
- Fast!

3D Object Recognition results

[Ferrari et al '04]

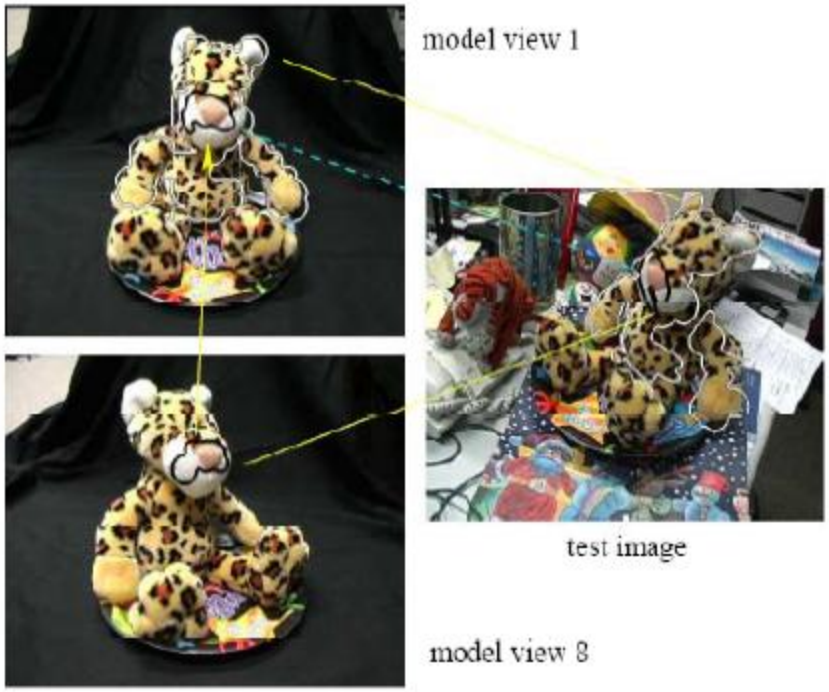


Figure 17: *Two compatible (and correct) GAMs. The nose GAM (black) is initially matched from model view 8, and is transferred to model view 1. Note how the other GAM (white) is very large and covers the head, arms and chest. A GAM can extend over multiple facets when the combination of viewpoints and surface orientations make the affine transformations of the region matches vary smoothly even across facet edges. In these cases, the resulting GAMs are larger and therefore more reliable and relevant.*



Courtesy of Ferrari et al

3D Object Recognition results

Edward Hsiao, Alvaro Collet and Martial Hebert. **Making specific features less discriminative to improve point-based 3D object recognition**. *IEEE International Conference on Computer Vision and Pattern Recognition (CVPR)*, June, 2010.



The end !